

AIIRLab at COLIEE 2026: Fusion of Instruction-Tuned LLMs and Cross-Encoders for Legal Case Entailment

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Abstract

The Competition on Legal Information Extraction and Entailment (COLIEE) provides a standardized benchmark for evaluating large language models across four distinct tasks: Legal Case Retrieval (Task 1), Legal Case Entailment (Task 2), Statute Retrieval (Task 3), and Legal Textual Entailment (Task 4). This paper presents the AIIR Lab’s submissions to COLIEE 2026, focusing on optimizing LLMs for the long-context and high-precision requirements of the legal domain. For Tasks 1 and 2, which operate over Federal Court of Canada case law, systems employ Parameter-Efficient Fine-Tuning (PEFT) via Low-Rank Adaptation (LoRA), bi-encoder dense retrieval, and multi-stage reranking pipelines. Task 2 introduces the Fusion3 ensemble, combining fine-tuned LLaMA-3.1-8B, Qwen2.5-7B, and a DeBERTa cross-encoder via union voting, which achieves an F1 of 0.47 and ranks second among all participating teams. For Tasks 3 and 4, which target the Japanese Civil Code, systems leverage Hypothetical Document Embeddings (HyDE), multilingual dense retrieval, and ensemble strategies including Neuro-Symbolic (NeuSym), Cross-Lingual Chain-of-Thought (CrossCoT), and Architecturally Diverse (DivVote) voting ensembles. Results across all tasks reveal a fundamental trade-off between broad-coverage retrieval and high-precision reasoning, suggesting that the next generation of legal AI must move beyond semantic similarity toward systems capable of reliably distinguishing general rules from their procedural exceptions.

CCS Concepts

• **Applied computing** → Law; • **Information systems** → *Specialized information retrieval*.

Keywords

Legal Case Retrieval, Legal Entailment, Legal Language Processing

1 Introduction

Legal information processing remains a challenging setting for modern natural language processing systems because relevant evidence is often distributed across long documents expressed in highly specialized language [14]. Although large language models (LLMs) and retrieval-augmented pipelines have substantially improved semantic matching in open-domain tasks, legal applications

continue to require much stricter evidence alignment and more reliable reasoning than topical similarity alone can provide [12, 17]. In particular, legal retrieval and entailment systems must often operate over lengthy judicial opinions and statutory provisions, where important signals may appear only in a few paragraphs or even in a single sentence.

The Competition on Legal Information Extraction and Entailment (COLIEE) [11, 13] provides a standardized benchmark for evaluating these capabilities across four distinct tasks: Legal Case Retrieval (Task 1), Legal Case Entailment (Task 2), Statute Retrieval (Task 3), and Legal Textual Entailment (Task 4). The Artificial Intelligence and Information Retrieval (AIIR) Lab from the University of Southern Maine participated in this competition, providing runs for all four tasks.

Tasks 1 and 2 operate over a corpus of Federal Court of Canada judicial decisions. Task 1 requires identifying noticed cases, cases explicitly cited in an original decision, from a corpus in which all such citations have been redacted. Task 2 proceeds further by requiring the identification of specific paragraphs within a noticed case that entail a given decision fragment, demanding fine-grained semantic matching and logical inference at the paragraph level. Tasks 3 and 4 shift the focus to the Japanese Civil Code. Task 3 jointly requires statute retrieval and binary entailment prediction, where a correct answer must be supported by an appropriately retrieved article. Task 4 isolates the entailment component by providing gold-standard relevant articles at test time, enabling a more direct evaluation of legal reasoning capabilities independent of retrieval performance.

Across all tasks, the systems presented in this paper implement a range of strategies for these challenges. For case-based tasks, these include Parameter-Efficient Fine-Tuning (PEFT) via Low-Rank Adaptation (LoRA), bi-encoder dense retrieval with hard-negative mining, and multi-stage reranking using both cross-encoders and generative LLMs. For statute-based tasks, strategies include Hypothetical Document Embeddings (HyDE), multilingual dense retrieval, pairwise LLM reranking, and ensemble voting architectures. For Task 2, the Fusion3 system, a union ensemble of fine-tuned LLaMA-3.1-8B, Qwen2.5-7B, and a DeBERTa cross-encoder, achieves an F1 of 0.47 and ranks second among all participating teams. For Task 4, three ensemble configurations are compared: a Neuro-Symbolic ensemble (NeuSym) that incorporates symbolic backward-chaining inference, a Cross-Lingual Chain-of-Thought ensemble (CrossCoT) that augments a multilingual baseline with QLoRA-trained chain-of-thought reasoning, and an Architecturally Diverse ensemble (DivVote) that introduces a BART-large sequence-to-sequence component to mitigate encoder-only bag-of-words failure modes.

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Table 1: Dataset statistics for training and evaluation splits.

Split	Queries	Corpus Size	Rel. Pairs	Avg. Rel./Query
Train	2,001	7,708	8,251	4.12
Test	400	1,848	1,750	4.38

Findings across tasks show a consistent tension between broad-coverage retrieval and high-precision reasoning. Generative rerankers improve semantic matching but face recall bottlenecks when critical precedents are buried in long documents or cut off by context-window constraints. Pointwise reranking pipelines achieve high recall but are susceptible to score saturation and semantic confusion between adjacent statutory provisions. Symbolic decomposition and chain-of-thought reasoning offer targeted improvements for nested exception handling and fine-grained qualitative distinctions, but their benefits are task- and instance-specific. The remainder of this paper is organized as follows: Sections 2 through 5 each describe one of the four tasks, presenting the task definition, proposed systems, and evaluation results in turn, followed by a concluding discussion in Section 6.

2 Task 1: Legal Case Retrieval

In this section, we first review the legal case retrieval task and then explain our proposed models, followed by evaluation results.

2.1 Task Definition

Given a query case q , the task is to retrieve a ranked list of noticed cases from a fixed corpus. A case is considered noticed if it is explicitly referenced in the original judicial decision. However, for a realistic retrieval setting, all such references are removed (redacted) from the query case text.

This task uses a corpus derived from Federal Court of Canada case law, where both query and candidate cases are provided as individual text files. The dataset is divided into:

- (1) *Training set*: Includes query cases along with annotated noticed cases, provided as a JSON mapping from query IDs to lists of relevant case IDs.
- (2) *Test set*: Contains only query cases; systems must predict the corresponding noticed cases.

System performance is evaluated using standard information retrieval metrics:

- (1) Precision: Proportion of retrieved cases that are relevant.
- (2) Recall: Proportion of relevant cases that are successfully retrieved.
- (3) F1-score: Harmonic mean of precision and recall.

Micro-averaged evaluation is adopted, where counts of true positives, false positives, and false negatives are aggregated across all queries before computing the metrics. The distribution of queries, corpus size, and relevance labels is summarized in Table 1.

2.2 Proposed Models

Preprocessing Step. Two of our systems, QWEN and LLaMA, use the same cleaning step. Raw case files from the Federal Court of

Canada were passed through a two-stage pre-processing pipeline before any downstream analysis. In the first, deterministic stage, each document underwent a sequence of rule-based cleaning steps; line endings were normalized, procedural headers appearing before the first numbered paragraph were stripped, and suppressed-fragment placeholders along with a curated list of bilingual boilerplate tokens (e.g., translation notices, emphasis markers, omission indicators) were removed. Because the corpus is officially bilingual, paragraphs whose sentences were in French, detected via a lexicon of common French function words and legal terms combined with a diacritic heuristic, were discarded, retaining only the English-language content. Remaining whitespace artifacts, punctuation, and excess blank lines were then normalized to produce clean, consistently formatted plain text.

In the second stage, each cleaned document was processed by the *LLaMA-3.1-8B-Instruct* [3] model to extract only legally important content. Because many case files exceeded the model’s context window, documents were split into overlapping chunks of approximately 6,000 characters with a 400-character overlap; boundaries were snapped to paragraph breaks to avoid mid-sentence cuts. Each chunk was submitted independently to the model with explicit positional context (chunk i of N), instructing it to retain legal reasoning, factual findings, statutory interpretation, and procedural history while discarding citation lists, counsel listings, and redundant headings. Chunks flagged by the model as containing no legally significant content were silently skipped. When a document produced multiple chunk extractions, a final LLM call merged them into a single coherent prose summary, eliminating redundancy introduced by the overlapping windows.

Our ParaNemo system utilizes a similar pre-processing strategy, but is single-staged and fully deterministic. The documents passed to the system are split into paragraphs using the ordered lists denoted by ‘[x]’ in each document in the dataset. Then, each paragraph is tokenized with spaCy¹ for the removal of extra whitespace, stop-words, and punctuation. The French language is not removed, translated, or summarized.

After pre-processing, we proposed three distinct systems as follows:

QWEN. The proposed system adopts a retrieve-then-rerank architecture. In the first stage, a bi-encoder model *all-mpnet-base-v2* [16] is fine-tuned on legal case pairs. Training pairs are constructed by splitting each case document into paragraphs and using BM25 to align passages from query cases with their known relevant (noticed) cases, yielding semantically grounded positive pairs at the passage level. The model is fine-tuned with Multiple Negatives Ranking loss and validated via embedding similarity on a held-out 10% split, with early stopping to prevent overfitting. At inference, each document is encoded at the paragraph level, and queries are matched against the entire corpus using a MaxSim scoring strategy, taking the maximum cosine similarity across all query-passage and candidate-passage pairs, which allows the model to reward partial but topical overlap without requiring full-document alignment. The top candidates from this stage are then passed to the re-ranking stage.

¹<https://spacy.io/>

In the second stage, a Qwen2.5-7B-Instruct [4] model acts as a pointwise relevance judge over the top-10 candidates retrieved per query. For fine-tuning, the model is adapted using QLoRA via Unsloth,² with LoRA adapters (rank 16, alpha 32) applied to all attention and feed-forward projection layers. Training data consists of positive (query, relevant case) pairs labeled with high confidence scores, and hard negatives mined using BM25 [15] to retrieve lexically similar but non-relevant cases, resulting in a 2:1 negative-to-positive ratio per query. The model is trained to produce a binary relevance judgment along with a continuous confidence score, using the same prompt format at both training and inference to eliminate distributional mismatch. At inference, both the pre-trained and fine-tuned Qwen models independently score each candidate, producing two ranked lists of relevant cases per query. These lists are then combined using CombMNZ, a data fusion method that rewards documents appearing in multiple result sets by multiplying their summed scores by the number of lists in which they appear. A fallback mechanism is used to ensure at least one result is returned per query in cases where the reranker rejects all candidates.

LLaMA. We propose a two-stage retrieve-and-rerank pipeline for the long-context requirements of the legal domain. In the first stage, a bi-encoder based on BGE-M3 [1] is employed for dense retrieval, selected for its support of input sequences up to 8,192 tokens. The model is fine-tuned on the COLIEE 2026 dataset using *Cached Multiple Negatives Ranking Loss* (InfoNCE), with five hard negatives per positive instance to improve discriminative capability. At inference time, query and document embeddings are precomputed, and candidates are ranked via cosine similarity; the top 50 documents per query are retained for downstream re-ranking.

In the second stage, LLaMA 3.1 8B Instruct [3] is used as a cross-encoder to refine the candidate set by framing re-ranking as a constrained generation task. The model is adapted using Parameter-Efficient Fine-Tuning with Low-Rank Adaptation (LoRA), with hyperparameters $r = 16$, $\alpha = 32$, and dropout 0.05. Training is performed via supervised token-level learning, prompting the model to act as a legal expert and produce a binary relevance judgment (“Yes”/“No”) indicating whether a candidate case supports the query. At inference, relevance scores are derived from the softmax-normalized logit of the “Yes” token. Candidates from the top-50 pool are re-scored, and those with $P(\text{Yes}) \geq 0.75$ are retained, subject to a final cutoff of five documents per query to prioritize precision.³

ParaNemo. The proposed system adopts a similar retrieve-then-rerank architecture as the QWEN system, but diverges slightly due to the differing pre-processing methods. In the retrieval stage, BM25 [15] is employed as the lexical baseline to retrieve an initial set of 100 candidate cases. The index is constructed by concatenating all paragraphs within each case from the preprocessed corpus. At query time, each preprocessed query paragraph is independently scored against the index. For a given query–case pair, scores across all query paragraphs are aggregated by taking the maximum value. The resulting document–score mappings are then ranked and forwarded to subsequent stages.

In the second stage, a fine-tuned bi-encoder (as used in the QWEN system) is used for semantic retrieval to re-rank and prune

Table 2: Results of AIIR Lab Runs for Task 1 on the COLIEE 2026 Test Set (400 Queries). * indicates corrected post-run.

Run Name	F1	Precision	Recall
QWEN	0.1516	0.2642	0.1063
ParaNemo	0.1125	0.1055	0.1206
LLaMA*	0.2489	0.2370	0.3396

the BM25 candidates to the top-10. Similarity is computed between query and case paragraphs using a pairwise similarity matrix over their preprocessed representations. For each query–case pair, the maximum paragraph-level similarity score is used as the aggregate relevance signal, based on the intuition that a case may be “noticed” due to a single highly relevant paragraph. Alternative aggregation strategies, such as mean and top-k mean pooling, were also considered. The resulting similarity scores are used to rank the candidates, and the top 10 are forwarded to the final re-ranking stage.

In the final stage, a cross-encoder, *llama-nemotron-rerank-1b-v2* [2], is used to re-rank the candidates produced by the bi-encoder. The model supports inputs of up to 8,192 tokens and is multilingual, which is advantageous given the characteristics of the corpus. Prior to re-ranking, all paragraphs within each preprocessed case are concatenated to reduce the number of inference calls and to better utilize the model’s input capacity. This design also allows the model to capture contextual dependencies that may span multiple paragraphs. The model supports prompt-based re-ranking, for which we use the following template:

case:{query}

potentially-relevant-case:{passage}

After re-ranking, the top 5 cases are retained as the final output to mitigate the impact of lower-ranked, potentially noisy candidates.

2.3 Evaluation Results

Table 2 shows our systems’ results for Task 1, with a post-run-corrected result from the LLaMA system. The LLaMA model achieves the highest F1 Score and recall, significantly higher than other models, using a paired Student’s t-test ($p < 0.05$). The QWEN model’s precision is significantly higher than the two other models, with a precision above 0.5 for nearly 30% of queries. However, this model had recall of 0 for 65% of queries, whereas LLaMA failed to retrieve relevant results for 35% of test queries.

Looking at the results, The performance gap between the two LLM-based systems shows how each model prioritizes different aspects of legal similarity. The LLaMA-based pipeline is more effective at retrieving relevant cases when the connection was based on broad doctrinal frameworks rather than specific statutory anchors. In Queries 000440 and 022674, for example, the relevant cases share overarching principles, such as the functional integration test for federal jurisdiction or requirements for representative standing, which are often expressed through varied terminology. In Query 000440 and Case 060605, the jurisdictional analysis is framed through very different factual backgrounds, making LLaMA’s document-wide alignment more successful than localized

²<https://github.com/unslothai/unsloth>

³During the competition, this system was run on a wrong input file, and the corrected run is reported in the evaluation section.

Table 3: Comparison of ParaNemo re-ranking versus the standalone fine-tuned bi-encoder candidate set.

Retrieval Pipeline Stage	F1	Precision	Recall
ParaNemo (Final Re-rank)	0.1125	0.1055	0.1206
Fine-tuned Bi-encoder (Top-10)	0.1534	0.1103	0.2520

matching. By using the 8,192-token context window of the BGE-M3 encoder, LLaMA captures reasoning patterns distributed throughout a judgment. This capability allows it to identify candidates like Query 076750, where the abuse of process doctrine is articulated through disparate procedural facts.

On the other hand, the Qwen-based ensemble achieved higher precision by prioritizing paragraph-level alignment. This approach effectively filters out the noise that can lead to false positives in document-wide models. The LLaMA pipeline’s use of character-level truncation and a high precision threshold of 0.75 makes it vulnerable to missing cases when key principles are cut off or buried in long factual narratives. Qwen’s success on Queries 025964, 080149, and 094642 instead stems from its MaxSim strategy and CombMNZ fusion, which isolate specific statutory points like Regulation 15 of the IRP Regulations or the Section 6(5) factors of the Trade-marks Act. By rewarding candidates that appear consistently across both pre-trained and fine-tuned lists, the Qwen ensemble maintains high precision for cases that require identifying narrow procedural links.

The performance of the ParaNemo system, summarized in Table 3, shows that the final cross-encoder re-ranking stage degraded retrieval metrics compared to the candidate set produced by the bi-encoder. Specifically, the fine-tuned bi-encoder achieved a higher F1 (0.1534) and Recall (0.2520) than the integrated ParaNemo pipeline. This suggests that while the *llama-nemotron-rerank-1b-v2* model provides a large 8,192-token context window and multilingual support, the transition from paragraph-level matching to full-document concatenation introduced new bottlenecks.

For instance, in query 028326, ParaNemo successfully retrieved the relevant Case 091015 by capturing dependencies across paragraph boundaries that the bi-encoder missed. However, this benefit was often outweighed by truncation issues. In query 032123, the re-ranker demoted a gold-label case that the bi-encoder had correctly identified. Because that document was exceptionally long (~65K characters), key legal points likely fell outside the 8,192-token limit, making them invisible to the cross-encoder while remaining accessible to the bi-encoder’s segment-wise matching. These results suggest that document-level concatenation can dilute the signal in long-form legal texts. The bi-encoder’s approach is more robust to document length because it isolates high-similarity segments, whereas the cross-encoder can be overwhelmed by factual noise or limited by its input capacity. Additionally, a mismatch exists between the English-centric bi-encoder and the multilingual cross-encoder, potentially causing the system to prune non-English candidates too early.

3 Task 2: Legal Case Entailment

This section reviews the legal case entailment task and provides explanations of our proposed systems and evaluation results.

3.1 Task Definition

While Task 1 focuses on identifying relevant (noticed) cases, this task aims to determine which specific parts of those cases justify a legal conclusion. Given a query case q , a noticed case c (assumed to be relevant to q), and a decision fragment d extracted from q , the goal is to identify one or more paragraphs in c that entail the decision fragment d . Unlike full case outcome prediction, the notion of “decision” here refers to a localized judicial conclusion (i.e., a fragment of reasoning), rather than the final verdict of the case. Systems must therefore perform semantic matching and logical inference between the decision fragment and candidate paragraphs. The dataset uses the same underlying corpus of Federal Court of Canada case law as Task 1, but is organized differently to support paragraph-level reasoning. The evaluation metrics are also the same as the ones used for Task 1.

3.2 Proposed Models

Fusion3. This model ensembles three different models as shown in Figure 1. Two instruction-tuned generative models are fine-tuned using QLoRA via Unsloth: *Meta-Llama-3.1-8B-Instruct* [3] and *Qwen2.5-7B-Instruct* [4], both loaded in 4-bit quantization. LoRA adapters with rank $r = 16$, $\alpha = 32$, and dropout 0.05 are attached to all attention and feed-forward projection layers.

Each model is trained for 3 epochs with a per-device batch size of 2 and 4 gradient accumulation steps (effective batch size of 8), a learning rate of 2×10^{-4} with a cosine schedule and 10% linear warmup, and early stopping with a patience of 5 evaluation steps. Training instances are constructed as three-turn chat conversations: a system message describing the task, a user turn presenting the decision fragment and candidate paragraph, and an assistant turn containing the binary label. The system prompt used is:

“You are a legal expert. Determine whether the PARAGRAPH below logically entails (supports / is the basis for) the DECISION FRAGMENT. Answer with exactly one word: ‘YES’ if it entails, or ‘NO’ if it does not.”

The user turn is structured as follows:

```
### Decision Fragment
<fragment truncated to 800 characters>

### Paragraph
<paragraph truncated to 1000 characters>

Does the paragraph entail the decision fragment?
```

Gold paragraphs are labeled YES and up to three randomly sampled non-relevant paragraphs per positive serve as hard negatives (labeled NO), yielding a 3:1 negative-to-positive ratio. At inference, rather than generating a token and thresholding its value, we perform a single forward pass and compute a continuous entailment score as $\log P(\text{YES}) - \log P(\text{NO})$ over the logits at the first generated position, taking the maximum log-probability across all single-token surface forms of each label (e.g., YES, yes, Yes, with and without a leading space).

The third component is a fine-tuned DeBERTa cross-encoder model [10] (described later in this section). At inference, the entailment class probability (softmax index 0 from the three-class NLI head) is used as the relevance score, and all paragraphs exceeding a threshold of 0.9 are returned as predictions, with a fallback to the

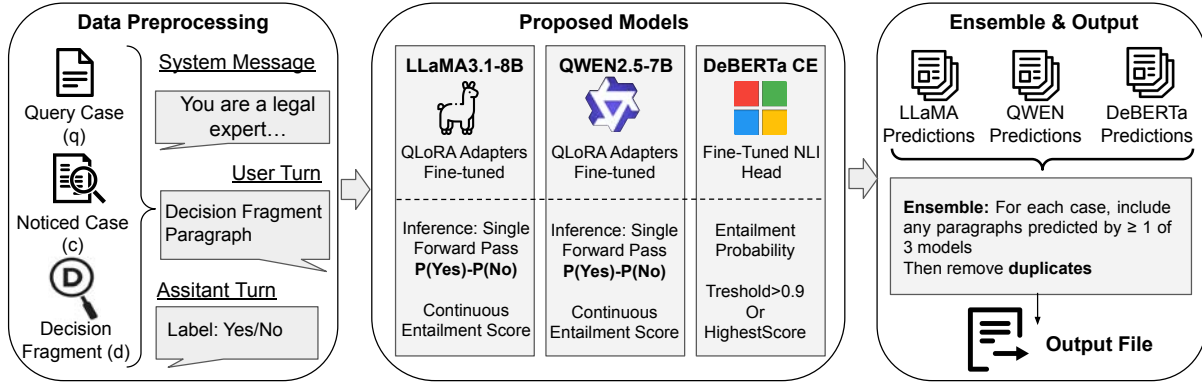


Figure 1: Fusion3 System Architecture. The task inputs (q,c,d) are processed, constructing three-turn chat instances with a 3:1 negative-to-positive ratio for fine-tuning. The Fusion3 ensemble consists of three models: fine-tuned Llama-3.1-8B and Qwen2.5-7B (each using QLoRA and 4-bit quantization), and a fine-tuned DeBERTa Cross-Encoder. The generative models compute a continuous entailment score via a log-probability difference, while DeBERTa uses a threshold of 0.9 with a highest-scoring fallback. The predictions from all three models are combined using a Union Ensemble, followed by a deduplication step to generate the final output.

highest-scoring paragraph when no candidate clears the threshold. The predictions from all three models are combined via a union ensemble: for each case, any paragraph predicted as entailing by *at least one* of the three models is included in the final output. Duplicate predictions across models are deduplicated, and results are written in the standard task submission format.

Fusion2. The proposed system utilizes a heterogeneous ensemble architecture specifically engineered to evaluate entailment between legal decisions and candidate supporting paragraphs. This pipeline processes every (decision, paragraph) pair in parallel across multiple GPUs using a late-fusion approach to aggregate model outputs. To capture complementary semantic features, the ensemble integrates two large language models with distinct training paradigms. The first component, Model 1, is a fine-tuned LLaMA3.1-8b-Instruct [3] model previously optimized via Low-Rank Adaptation (LoRA) for case retrieval tasks. This model is repurposed for entailment through specialized prompt framing that instructs it to act as a legal expert, performing binary classification by answering whether a paragraph directly supports a legal decision. The resulting relevance score is derived from the logit difference of the ‘Yes’ and ‘No’ tokens at the final sequence position.

To provide a robust baseline and mitigate over-optimization, Model 2 incorporates Qwen3-14B [5] in a zero-shot configuration. To ensure deterministic classification and immediate response, Chain-of-Thought (CoT) reasoning is explicitly disabled by injecting a closed-tag sequence (`<think>\n\n</think>\n`) immediately following the assistant turn. This forces the model to generate the classification token as the initial output, from which the $P(\text{Yes})$ logit-ratio is similarly extracted.

To account for the differing logit scales and distributions of the two models, we implement a normalized fusion strategy. During parallel scoring, both models independently evaluate all candidate paragraphs on separate GPUs. For each case, the raw score vector (s) for each model is mapped to the range $[0, 1]$ using min-max scaling for a specific paragraph p . The final ensemble score is then

calculated as the unweighted mean of these normalized probabilities. Paragraphs meeting a threshold of $P \geq 0.75$ are selected for the final submission. In instances where no paragraph clears this threshold, the system defaults to a ‘Top-1’ fallback strategy, returning the single highest-ranked paragraph to ensure a minimum result set and complete coverage.

Deberta. We fine-tune a DeBERTa-v3-base[10] cross-encoder to identify which paragraph(s) within a noticed case entail a given decision fragment from the query case. The model is initialized from a pre-trained three-class NLI checkpoint (entailment / neutral / contradiction) and fine-tuned end-to-end on case-specific paragraph pairs, where gold paragraphs are labeled as entailment and randomly sampled non-relevant paragraphs serve as negatives at a 5:1 ratio. Each training instance is formed by truncating the decision fragment to 200 words and the candidate paragraph to 280 words, keeping the combined input within the model’s 512-token limit. The model is trained for up to 40 epochs using a cosine learning rate schedule with 10% linear warm-up, and the best checkpoint is selected by MRR@10 on a held-out 10% validation split evaluated via a custom entailment re-ranking evaluator. At inference, raw logits from the cross-encoder are converted to probabilities via softmax, and the entailment class probability (index 0) is used as the continuous relevance score; all paragraphs exceeding a confidence threshold of 0.9 are selected as predictions, with a fallback to the highest-scoring paragraph when no candidate clears the threshold.

3.3 Evaluation Results

Table 4 shows our systems’ performance for Task 2. Fusion3 run provided the highest effectiveness, and was ranked as the second run in the COLIEE 2026 among the participating teams. Fusion3 achieved the highest F1 score (0.48), significantly outperforming Fusion2 (F1=0.42) and DeBERTa (F1=0.35) based on a paired Student’s t-test ($p < 0.05$). Fusion3 had a precision above 0.5 for 70%

Table 4: Results of AIIR Lab Runs for Task 2 on the COLIEE 2026 Test Set (100 queries).

Run	F1	Precision	Recall
Fusion3	0.4706	0.5120	0.4354
Fusion2	0.4271	0.4257	0.4286
Deberta	0.3333	0.5750	0.2347

of queries and a recall above 0.5 for nearly half of the queries. DeBERTa model, due to its strict thresholding policy, provided the highest precision among our runs, with a precision of 1 for more than half of the queries.

Fusion3’s results indicate that the union ensemble of instruction-tuned generative models combined with the DeBERTa-v3-base[10] cross-encoder provides a more comprehensive coverage of the entailment space. While the performance of Fusion2, which utilized a normalized fusion of LLaMA3.1-8b-Instruct[3] and Qwen3-14B[5], resulted in a precision score significantly lower than Fusion3’s, but was able to maintain a similar recall level. This gap in scores may be attributed to the differing ensemble strategies where Fusion3’s union approach of including any prediction appears better suited for the task’s retrieval requirements rather than the unweighted mean of normalized probabilities used in Fusion2.

As shown when analyzing the submission for query with ID 947, where the ground truth includes paragraphs 010, 011, Fusion2 returned a single result of 011 against Fusion3’s result: 007, 010, 011. Both systems returned 011, which satisfies the clearly stated issues of “CAIPS notes” and the applicant’s appeal on “humanitarian and compassionate considerations” for residency in Canada. However, Fusion3 also returned 010 which addresses the more subtle issue of “the best interests of the child” raised, and cites “Baker v. Canada (Minister of Citizenship and Immigration), [1999] 2 S.C.R. 817” which is also cited in the query. The citation of “Baker v. Canada (Minister of Citizenship and Immigration), [1999] 2 S.C.R. 817” is also present in 007, which would provide an explanation for why it, too, was returned by the Fusion3 system’s union approach.

The Fusion3 model’s zero-precision failures on these instances can be attributed to two main issues. First, the union ensemble’s design trades precision for recall; when any component model incorrectly predicts entailment due to shallow keyword overlap, erroneous paragraphs propagate to the final output unchecked. Second, the 800–1000 character truncation limits combined with the capacity of 7–8B parameter models to perform single-forward-pass entailment judgment are insufficient for the complex legal reasoning these queries demand. For instance, Query 937 requires weighing conflicting quantitative valuations of a seized vessel across multiple evidence sources, and Query 987 demands verification of a three-part cumulative confidentiality test; multi-step dependencies that small generative models struggle to resolve. Similarly, Queries 950 and 980 require normative and procedural inference (e.g., applying the *Khalil* exception or distinguishing policy categories for migrant healthcare), causing models to retrieve topically related but logically insufficient paragraphs.

4 Task 3: Statute Retrieval

This section provides details on task 3, providing details on the task, our proposed systems along with the evaluation results.

4.1 Task Definition

This task focuses on the joint problem of statute retrieval and textual entailment for legal question answering. Given a natural language query, the objective is to predict a binary decision (Yes/No) and retrieve a set of relevant statute articles that justify the answer. An article is considered relevant if its content entails the answer to the query. As such, the task requires systems to perform integrated reasoning and retrieval, where correct predictions must be supported by appropriate legal evidence rather than relying solely on classification.

The corpus used in this task is derived from Japanese civil law and follows a structure similar to Tasks 1 and 2, in that relevance relationships are explicitly provided in the training data. Each instance consists of a query paired with one or more statute articles and an associated entailment label. The statutory texts are provided in Japanese, and participants may optionally incorporate machine translation or multilingual modeling techniques. The training data includes both the relevant articles and the correct entailment labels; whereas the test data contains only queries. Systems must therefore automatically retrieve the relevant articles and infer the correct Yes/No decision without any human intervention or test-time adjustments.

Evaluation emphasizes both correctness and completeness of the predicted outputs. The primary metric is accuracy, defined as the proportion of queries for which the system predicts the correct entailment label and retrieves a sufficient set of relevant articles (i.e., achieving full recall for that query). In cases of ties, the F2-measure is used as a secondary metric, placing greater emphasis on recall over precision.

4.2 Proposed Models

JPGen. This system implements a Hypothetical Document Embedding (HyDE) pipeline that decomposes the retrieval and entailment task into two sequential stages: article generation followed by dense retrieval. The intuition behind this approach is that a query expressed in natural language may be semantically distant from the formal statutory language of the corpus, and generating an intermediate “hypothetical article” can bridge this gap by projecting the query into the distributional space of legal text.

In the first stage, a *Qwen2.5-7B-Instruct* [4] model is fine-tuned using Low-Rank Adaptation (LoRA) on query–article pairs drawn from the training set. The model is trained to generate a relevant statutory article given a legal query, using the system prompt: “You are a Japanese legal expert. Given a statement related to law, generate a legal article that supports or is related to it. Output only the article text, without any explanation or preamble.” Training uses 4-bit quantization via the Unsloth framework, with a LoRA rank of 16 and a learning rate of 2×10^{-4} , and a cosine learning rate schedule over three epochs. An early stopping callback with a patience of five evaluation steps is applied to prevent overfitting, with the best checkpoint selected according to validation loss.

In the second stage, the generated articles are used as the retrieval query against the full statute corpus. Candidate articles are ranked using a fine-tuned bi-encoder based on *multilingual-e5-large* [8], and the top-5 retrieved articles are returned as the predicted relevant set. The entailment label is subsequently determined from the retrieved evidence.

JPLLM. This system uses a three pipeline that narrows the article collection down to a final set of relevant statutes. The first stage employs a fine-tuned bi-encoder based on *multilingual-e5-large* [8], a multilingual model that natively supports Japanese. Following the E5 training convention, all queries are prefixed with "query: " and all article passages are prefixed with "passage: " at both fine-tuning and inference time. The model is fine-tuned using Multiple Negatives Ranking Loss (MNRL) on positive (query, article) pairs extracted from the training labels, with in-batch negatives as the contrastive signal, for 20 epochs with a batch size of 8 and 10% linear warmup. Validation is performed using MRR@10 and NDCG@10 on a held-out 10% split, and the best checkpoint is retained. At inference, the entire article collection is encoded once and each query is matched against it via cosine similarity (implemented as a dot product over L2-normalised embeddings), retrieving the top-50 candidates per query.

The top-50 candidates from Stage 1 are re-ranked using a fine-tuned *Qwen2.5-7B-Instruct* [4] model operating as a pairwise comparator. The model is fine-tuned on the training set as a binary classifier to detect whether a given article is relevant to a query. At inference, for each query the model is presented with two candidate articles and asked to identify the more relevant one, responding with a single digit (1 or 2). The pairwise comparisons are structured as a selection-sort procedure: k passes are made over the candidate pool, each identifying the best remaining article via pairwise LLM calls, producing the top-20 ranked articles per query. All prompts are issued in Japanese to avoid language-switching overhead; the system prompt (translated from Japanese) instructs the model as follows:

"You are a legal document re-ranking system. Given a legal query and two articles, decide which article is more directly relevant to the query. Reply with only a single digit: 1 if Article 1 is more relevant, or 2 if Article 2 is equally or more relevant."

The user turn presents the query followed by the two candidate articles and asks the model to respond with 1 or 2. Up to three retries are performed on ambiguous outputs before defaulting to no swap.

The top-20 candidates from Stage 2 are passed through a final point-wise relevance filter using the same *Qwen2.5-7B-Instruct* [4] model. For each (query, article) pair, the model outputs a continuous relevance score in [0, 1] on the first line, followed by a one-sentence justification on the second line. Prompts are again issued in Japanese; the system prompt (translated from Japanese) applies a strict relevance criterion:

"You are a strict legal relevance assessor in a civil law information retrieval system. An article is considered relevant only if it directly and specifically addresses the legal issue raised in the query. Articles that merely reference adjacent legal concepts or are only indirectly

applicable should be judged as not relevant. Assign a relevance score between 0.0 and 1.0, where 1.0 means the article directly and completely addresses the query, and 0.0 means it is entirely unrelated. Apply a high relevance threshold: scores above 0.5 should be reserved only for articles that directly address the specific legal issue in the query. If the judgment is difficult, assign a score below 0.5."

Articles with a predicted score above 0.5 are retained as relevant; if no article clears this threshold, the highest-scoring article is included as a fallback to guarantee at least one result per query.

For the entailment sub-task, the articles retrieved by the three-stage pipeline are passed to the fine-tuned *Qwen2.5-7B-Instruct* [4] model, which makes a binary entailment decision, determining whether the legal statement in the query is entailed (Y) or not (N) given the retrieved articles as supporting evidence.

JPSW. This system follows a three-stage pipeline specifically optimized for the Japanese Civil Code corpus. The system transitions from broad multilingual dense retrieval to Ensemble re-ranking to generative re-ranking and logical reasoning.

Stage 1: Multilingual Dense Retrieval. The initial retrieval phase utilizes the *multilingual-e5-large-instruct* [8] model to perform dense semantic search. The 800-article Civil Code corpus is pre-computed into a latent vector space and stored as high-dimensional embeddings. During inference, each query is prepended with a task-specific instruction:

"Instruct: Given a legal question about Japanese civil law, retrieve the most relevant civil code article. Query: [User Query]".

Candidates are ranked using cosine similarity, and the top-100 most relevant articles are surfaced for the subsequent stages.

Stage 2: Ensemble Re-ranking. The candidate pool is refined through an ensemble of Large Language Models, consisting of *Llama-3.1-8B-Instruct* [3], *Llama-3.1-Swallow-8B-Instruct* [6], and *Gemma-2-Llama-Swallow-9b* [6]. All three models were fine-tuned using Low-Rank Adaptation (LoRA) to handle Japanese legal syntax. This ensemble utilizes a dual-task prompt framing where the model must provide both a relevant article number and a binary (Yes/No) entailment verdict for a given article-statement pair. The scores from these three models are aggregated to prune the candidate set from 100 to the top-50 most probable articles.

Stage 3: Generative Re-ranking & Entailment. The top-50 candidates are further re-scored by a larger *GPT-OSS-Swallow-20B* [6] model. This stage performs a more granular evaluation of the legal nuances within the top-50 pool, reducing the result set to a final selection of K candidates (where $K = 10$ by default). For Entailment, Chain-of-Thought processing is employed to improve logical consistency, and any reasoning is stripped to extract the binary classification. The final retrieval scores undergo global normalization across all queries and a dynamic thresholding logic is applied. If the top-ranked candidate's normalized score exceeds 0.6, all candidates with a score greater than or equal to 0.6 are retained. If the primary candidate does not meet the 0.6 threshold, the system defaults to a conservative fallback strategy, returning only the rank-1 result.

4.3 Evaluation Results

Table 5 shows our systems’ results for Task 3. As shown in this table, JPGEN and JPLLM both have significantly higher recall compared to JPSW, while JPSW provides a significantly higher precision (Student’s paired t-test $p < 0.05$). On one hand, JPSW for more than half of the queries provides a precision higher than 0.5. On the other hand, JPLLM has a recall above (0.5) for more than 80% of queries.

The queries for which JPSW achieves perfect precision while JPLLM retrieves no relevant articles (e.g., R07-12-O, R07-14-A, R07-23-E, and R07-28-U), each is grounded in exactly one relevant statute, covering legally self-contained topics such as creditor expense recovery, mortgage extinction, novation with mortgage, and a partnership member’s disposal of their share. For each of these queries, JPSW returned a single article assigned a high normalised relevance score ($0.829 \cdot 1.0$), correctly matching the ground truth. JPLLM, by contrast, assigned the maximum score of 1.0 to every article in its five-item output, none of which were relevant. This score saturation pattern in JPLLM’s pointwise re-ranking stage, where the Qwen model’s confidence no longer discriminates meaningfully among candidates, is caused by its fallback mechanism, which guarantees at least one returned result per query. For these queries, JPLLM suffers from domain-level semantic confusion; for R07-23-E (article 518, novation with mortgage), JPLLM returns articles from the assignment and novation chapter immediately surrounding the correct provision, and for R07-28-U (article 676, partnership share disposition), it returns articles 256–260 from the co-ownership chapter entirely. The overlap between “share in partnership property” and “share in co-owned property” mislead the pair-wise re-ranker, while JPSW’s generative stage correctly resolves the distinction through its conservative thresholding strategy.

In cases where JPLLM achieves full recall while JPSW retrieves nothing relevant, JPSW either returns no results or commits to a single low-confidence article that is numerically or thematically adjacent to the correct provision. For R07-03-A (article 98-2, expressions of intent to a party under curatorship), JPSW returns article 97, the general notice-of-intent provision from which 98-2 departs as a specific exception; for R07-04-I (article 114, ratification silence in unauthorised agency), it retrieves article 113, the adjacent provision on demanding confirmation; and for R07-21-I, it returns sub-provision 466-6 when the query implicates the parent article 466. In all these cases, the system correctly identifies the legal chapter but cannot resolve the boundary between a provision and its procedural exception or sub-provision. JPLLM navigates these distinctions more reliably, placing the correct article within the top two ranks in the majority of these queries, confirming that the relevant provision is reachable by broad retrieval but is discarded by JPSW’s aggressive pruning. Across both systems, the primary retrieval challenge is therefore not broad domain identification but the fine-grained disambiguation of closely related provisions addressing subtly different legal scenarios.

5 Task 4: Legal Textual Entailment

In this section, we explore task 4, starting with the task definition, followed by our proposed systems and the evaluation results.

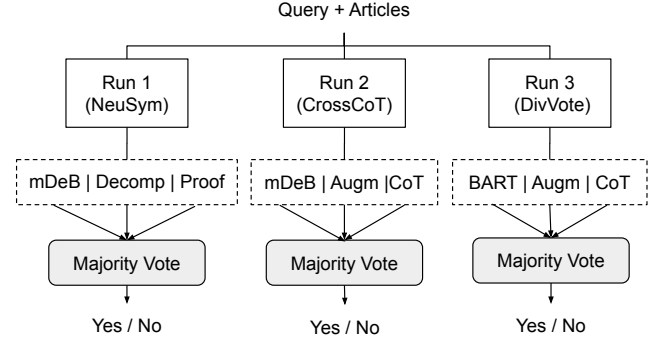


Figure 2: Overview of the three ensemble systems for Task 4. Each run consists of three complementary components whose predictions are aggregated via majority voting.

5.1 Task Definition

This task focuses on legal textual entailment for Yes/No question answering, assuming that the relevant statutory evidence is already provided. Given a natural language query and a set of golden (manually selected) relevant statute articles, the objective is to determine whether the query is entailed (Yes) or not entailed (No) by the provided articles. In contrast to Task 3, where systems must jointly retrieve and reason over statutes, this task isolates the entailment component, enabling a more direct evaluation of a system’s reasoning capabilities when the supporting evidence is known.

The corpus used in this task is the same as in Task 3, consisting of Japanese civil law articles paired with query statements in a structured format. Each instance links a query with one or more relevant articles and an associated Yes/No label indicating whether the query is entailed by those articles. The training data provides both the relevant articles and the corresponding entailment labels. At test time, systems are given queries together with the golden relevant articles, and must predict the correct Yes/No decision. As in the other tasks, the entire process must be fully automatic, with no human intervention or modifications based on inspection of the test queries.

Evaluation in this task focuses solely on the correctness of the entailment decision. The primary metric is accuracy, defined as the proportion of queries for which the system correctly predicts the Yes/No label. Unlike Task 3, retrieval quality is not evaluated, as the relevant articles are provided. This setup allows for a more controlled assessment of legal reasoning and entailment, independent of retrieval performance.

5.2 Proposed Models

We submitted three systems to COLIEE 2026 Task 4, each implemented as a three-member majority-vote ensemble as shown in Figure 2. All components are trained using a stratified 90/10 train-validation split of the provided Japanese data, with checkpoint selection based on validation macro-F1. Across all runs, we maintain a shared mDeBERTa cross-encoder baseline and introduce complementary components to capture structured reasoning, data augmentation effects, chain-of-thought (CoT) reasoning, and architectural diversity.

Table 5: AIIRLab Task 3 submission results for Accuracy, Correctly Retrieved, Sufficiently Retrieved, Accuracy of Sufficiently Retrieved, F2, Precision and Recall.

Run	Accuracy	Correct	Sufficiently Retrieved	Acc. Suff. Retrieved	F2	Precision	Recall
JPGEN	0.5854	48	63	0.7619	0.4481	0.1707	0.7785
JPLLM	0.5732	47	64	0.7344	0.4606	0.1770	0.7947
JPSW	0.5000	41	46	0.8913	0.5474	0.5183	0.5915

Shared Baseline: mDeBERTa Cross-Encoder. All ensembles include a fine-tuned *mDeBERTa-v3-base-mnli-xnli* [7] model as a core voter. The model takes the concatenated (article, query) pair as a standard NLI input and is trained with AdamW (learning rate 2×10^{-5} , weight decay 0.01), 10% linear warmup, and gradient clipping at 1.0. Mixed-precision training and gradient accumulation (4 steps) yield an effective batch size of 32. At inference, the entailment and contradiction logits are compared, predicting *Yes* when the former is larger. Training proceeds for up to 10 epochs, and the best checkpoint is selected by validation macro-F1.

Run 1: Neuro-Symbolic Ensemble (NeuSym). This ensemble augments the neural baseline with two structured reasoning components:

First, a *condition-conclusion decomposition model* converts each article and query into a structured *LegalStructure* representation (conditions, effect, exceptions, negation) using *Qwen2.5-7B-Instruct* [4] with LoRA 4-bit quantization. A hybrid model then combines (i) similarity scores computed via a fine-tuned *multilingual-e5-small* encoder [8] over decomposed fields, and (ii) the mDeBERTa entailment logit. These signals are fused through a learned interpolation weight, with separate learning rates for encoder and structural components.

Second, a *neuro-symbolic proof model* translates inputs into logical predicates and rules, performs backward-chaining inference using pyDatalog⁴, and extracts proof-derived features (e.g., proof success, depth, unresolved goals). These features are combined with the mDeBERTa score via a lightweight fusion module. This component provides interpretable reasoning traces while injecting symbolic structure into the decision process. The final prediction is obtained by majority voting across the baseline, decomposition, and proof-based components.

Run 2: Cross-Lingual Chain-of-Thought Ensemble (Cross-CoT). This ensemble focuses on improving generalization through data augmentation and explicit reasoning supervision.

The first additional component is a *data-augmented mDeBERTa model*. Synthetic training pairs are generated using *Qwen2.5-7B-Instruct* [4] under four strategies: paraphrasing and negation for *Yes* instances, and condition substitution and article-driven claim generation for *No* instances. Augmented samples are filtered to prevent validation leakage, and the model is trained identically to the baseline.

The second component is a *QLoRA fine-tuned CoT model*. Chain-of-thought rationales are generated in Japanese following a structured legal reasoning template, and only samples with correct final answers are retained. The base Qwen model is then fine-tuned using QLoRA via Unsloth for three epochs. At inference, the model

Table 6: AIIRLab Task 4 Results on COLIEE 2026 test set (82 Queries).

Run	Correct	Accuracy
CrossCoT	64	0.7805
DivVote	64	0.7805
NeuSym	57	0.6951

generates a reasoning trace and extracts the final Yes/No decision using pattern matching. Final predictions are obtained via majority voting across the baseline, augmented cross-encoder, and CoT model.

Run 3: Architecturally Diverse Ensemble (DivVote). This ensemble introduces architectural diversity while reusing strong components from Run 2. The first component replaces the baseline with a fine-tuned *facebook/bart-large-MNLI* [9] cross-encoder. Despite being pretrained on English, BART provides a complementary inductive bias as a sequence-to-sequence model. Training follows the same NLI formulation, with adjusted batch size and learning rate to accommodate model scale. The remaining two components, the data-augmented mDeBERTa model and the QLoRA CoT model, are identical to those in Run 2. As in the other runs, predictions are combined through majority voting.

5.3 Evaluation Results

Table 6 shows the evaluation results of our three runs on COLIEE 2026 test set. All three models had correct predictions on 50 instances, and on 9 queries all models provided wrong predictions. Both CrossCoT and DivVote provided the same number of correct instances with an accuracy of 0.78. However, these models had different predictions for 14 instances, for half of which each system had a correct prediction. The NeuSym system had lower accuracy, but for two queries it provided the correct prediction while the other two models failed.

The Neuro-Symbolic Ensemble (NeuSym) showed an advantage in handling complex condition-exception structures and taxonomic mappings. For instance, NeuSym correctly predicted entailment for instances involving nested exceptions, such as the guarantor exception clauses outlined in Article 504 (R07-22-O). Its *LegalStructure* decomposition isolates exceptions into distinct structural fields, preventing them from being overshadowed by the broader semantic overlap of general rules. Furthermore, NeuSym’s symbolic component succeeded in taxonomic mapping, such as properly categorizing a “real estate pledge” under the broader “real rights regarding real estate” in Article 177 (R07-08-U), a task where standard cross-encoders failed to establish sufficient dense semantic similarity.

⁴<https://github.com/pcarbonn/pyDatalog>

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The divergence between the Architecturally Diverse Ensemble (DivVote) and the Cross-Lingual Chain-of-Thought Ensemble (CrossCoT) shows the trade-offs between sequence-to-sequence and encoder-only architectures. DivVote correctly identified structural inversions and hierarchical priority orderings where CrossCoT failed, such as reversing the fault attribution between a creditor and debtor under Article 413-2 (R07-17-U) or falsely inverting the statutory lien priorities of funeral and daily necessity expenses under Article 306/329 (R07-12-A). By incorporating a *BART-large-MNLI* [9] cross-encoder, DivVote exhibited a stricter inductive bias for structural alignment, effectively avoiding the “bag-of-words” similarity traps that often mislead encoder-only models on highly overlapping adversarial texts.

However, DivVote’s reliance on an English-pretrained model introduced a semantic bottleneck when processing Japanese legal vocabulary, an area where CrossCoT’s native multilingual model excelled. CrossCoT correctly rejected false premises relying on lexical distinctions, such as conflating the mere “loss” of an item with the strict legal threshold of “deprivation” required for possessory actions under Article 200 (R07-10-A). Additionally, CrossCoT successfully anchored its attention to precise parameter checks, including specific quantitative alterations to the statute of limitations for bodily harm (R07-30-O) and identifying the correct party authorized to transfer a mortgage during a novation (R07-23-E). This underscores the necessity of a native multilingual baseline, combined with explicit CoT reasoning, to resolve fine-grained qualitative and quantitative mismatches in original legal texts.

6 Conclusion

This paper presents the AIIR Lab’s submissions to COLIEE 2026 across four legal information processing tasks, showing that multi-stage ensemble architectures are well-suited for the complex demands of legal retrieval and reasoning. In the case-based tasks, Fusion3’s union ensemble of fine-tuned LLaMA-3.1-8B, Qwen2.5-7B, and DeBERTa achieved an F1 of 0.47 in Task 2, ranking second among all participating teams, while the LLaMA pipeline’s document-wide alignment outperformed paragraph-level approaches in Task 1 recall. In the statute-based tasks, JPLLM’s pointwise reranking achieved recall up to 0.79 in Task 3 but was vulnerable to score saturation; in contrast, JPSW achieved significantly higher precision ($p < 0.05$), correctly resolving distinctions between complex legal topics such as partnership shares and co-owned property. In Task 4, results showed that architectural diversity and structured reasoning provide complementary gains, with CrossCoT and DivVote both reaching 0.78 accuracy and NeuSym uniquely resolving nested exception structures that purely neural models could not handle.

Our results suggest that the primary bottleneck in legal AI is no longer broad domain identification but fine-grained disambiguation; distinguishing a general rule from its procedural exceptions, a

provision from its sub-provisions, and topical relevance from logical entailment. Future work should focus on retrieval architectures that preserve paragraph-level granularity at scale and reasoning models capable of reliably resolving hierarchical legal structures.

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